MSBA Capstone Project

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A FourSight Analytics Designed Analytics Solution Proposal for Crafted Cones

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Executive Summary

Crafted Cones is a small, community-focused ice cream shop located in Oceanside, New York, that experiences strong summer demand but faces operational challenges due to seasonality and reliance on manual data processes. Currently, the business tracks sales, inventory, staffing, and customer interactions through fragmented spreadsheets and handwritten logs, which limits its ability to forecast demand, manage resources efficiently, and make data-driven decisions.

This project proposes a three-phase analytics roadmap designed to transition Crafted Cones from manual operations to a structured, analytics-enabled operating model supported by Customer Relationship Management (CRM) and Enterprise Resource Planning (ERP) systems. The solution includes centralized data capture, interactive Power BI dashboards, and time-series forecasting models that provide actionable insights across three strategic pillars: operations, workforce, and market & customer analytics.

Using a multi-year synthetic dataset that simulates the progression from manual recordkeeping to structured business systems, exploratory analysis and forecasting models were developed to evaluate seasonal demand patterns and operational performance. The analysis demonstrates strong seasonality in sales, with baseline forecasting models performing more reliably than advanced models due to limited historical data. As additional structured data becomes available, more advanced predictive models are expected to improve forecasting accuracy.

Financial analysis indicates that implementing analytics-driven operational improvements could generate approximately **$503,400 in annual cost savings** through improved scheduling, reduced employee turnover, lower product waste, and more targeted marketing promotions.

Overall, this project demonstrates how analytics can transform Crafted Cones into a data-driven organization capable of improving operational efficiency, strengthening customer engagement, and supporting strategic growth decisions. The recommended analytics roadmap provides a scalable framework for integrating data systems, generating actionable insights, and enabling long-term business expansion.

Company Overview

Crafted Cones is a locally beloved ice cream shop based in Oceanside, New York, known for its strong ties to the coastal community and its authentic, small-batch approach to frozen desserts. As a year-round business operating in a highly seasonal market, the company experiences peak demand during the summer months while navigating lower traffic and operational complexity throughout the colder seasons. This unique environment creates challenges in forecasting, production planning, staffing, and inventory management, particularly as the business has historically relied on manual spreadsheets, handwritten logs, and informal processes for daily operations.

The shop's brand identity is deeply rooted in local culture, community engagement, and flavor creativity, resulting in strong customer loyalty and a clear point of differentiation in a competitive market. By engaging actively with residents, incorporating local influences, and maintaining a warm neighborhood presence, Crafted Cones has cultivated a loyal customer base that values authenticity and connection.

The business faces operational limitations that hinder data-driven decision-making. Manual data capture, static reporting, and the absence of integrated systems restrict visibility into trends, slow responses to market changes, and limit the ability to plan effectively for seasonal fluctuations. These operational constraints highlight the need for a structured analytics framework that can transform fragmented operational data into actionable insights.

Overall, Crafted Cones is at an inflection point: a strong local brand with a loyal community following and a clear need for modernized systems that will enable data-driven strategic planning. This transformation will allow the business to maintain its community-centric identity while strengthening operational maturity and unlocking long-term growth.

Team and Project

Project Description

Crafted Cones faces several operational and strategic challenges stemming from its reliance on manual, unstructured processes for capturing sales, inventory movement, labor hours, customer interactions, and financial data. These manual approaches restrict the company's ability to understand real-time performance, anticipate changes in demand, or make proactive decisions; particularly in a business where seasonality, weather fluctuations, and community patterns significantly influence customer traffic and product needs.

The core objective of this analytics project is to help Crafted Cones transition to a more structured, data-informed operating model. The project focuses on identifying business opportunities, operational gaps, and areas where analytics can materially enhance decision-making across sales forecasting,

inventory management, workforce planning, and customer engagement. By analyzing current processes, identifying data needs, and outlining actionable insights supported by analytics, the project provides Crafted Cones with a clearer understanding of what drives performance and where efficiency and growth opportunities exist.

This work directly aligns with the company's broader strategic goals, including strengthening operational stability year-round and improving visibility into daily and seasonal trends. The project aims to provide solutions that enable Crafted Cones to evaluate factors such as seasonal sales patterns, production demands, staffing cycles, customer preferences, and inventory requirements. It also aims to craft a solution that highlights how external influences—such as weather, tourism, and local events—affect demand and should be incorporated into future planning.

Ultimately, this project serves as the foundational step in helping Crafted Cones develop a more analytics-enabled approach to operations. It outlines how transitioning from manual data entry to integrated CRM (Customer Relationship Management) and ERP (Enterprise Resource Planning) systems—supported by real-time dashboards and business intelligence tools—can translate operational data into accessible insights. These capabilities will support more accurate forecasting, more efficient resource allocation, and more dynamic strategic decisions.

Team Name and Organization

FourSight Analytics[1] is a women-led business analytics team that specializes in helping small, seasonal, community-focused businesses modernize their operations with simple, intelligent data systems. The four women who make up FourSight Analytics have worked with many small businesses facing challenges like heavy manual processes, unpredictable demand, and limited visibility into day-to-day performance.

Hannah Schroeder

Hannah Schroeder is a business analyst consultant specializing in data-driven strategy, product development, and stakeholder engagement. Currently completing a Master of Science in Business Analytics, she brings a proven track record of managing complex, multi-stakeholder projects and translating data insights into actionable business outcomes.

Hannah has successfully led concurrent project initiatives involving vendor coordination, cross-functional team collaboration, and end-to-end product lifecycle management. She excels at conducting stakeholder research, analyzing datasets to inform strategic decisions, and developing frameworks that drive organizational efficiency and profitability. She combines analytical rigor with innovative problem-solving to deliver results for consulting clients. Her core competencies include project management, data analysis, strategic planning, market research, and product roadmapping across diverse industries.

[1] FourSight Analytics is a fictitious business analytics group created for the purpose of this capstone project.

Breanna Rai

Breanna Rai is a business analyst specializing in performance measurement, workforce analytics, and data-driven strategy. She has experience designing evaluation frameworks, transforming complex operational data into actionable insights, and supporting senior leadership through evidence-based decision-making.

Her work includes leading analytics initiatives within Human Resources, such as survey design and interpretation, standardized reporting development, and the implementation of data governance practices. She integrates advanced Excel, SQL, and statistical analysis to improve organizational effectiveness and inform strategic planning. She is passionate about data communication and building analytical systems that empower organizations to make informed, strategic decisions.

Carrie Feltkamp

Carrie Feltkamp is an operations and strategy leader with a strong engineering foundation that informs her analytical, systems-oriented approach to problem-solving. With expertise spanning operational excellence, data analysis, and cross-functional leadership, she is skilled at translating quantitative insights into clear, measurable strategies that elevate organizational performance.

Carrie excels at simplifying complex workflows, building structures that enable consistent execution, and driving alignment through clear communication and collaborative partnership. While highly relationship-oriented, her leadership is rooted in disciplined analysis, process rigor, and an ability to connect metrics to meaningful business outcomes—ensuring that strategies are both people-supported and analytically sound.

Michelle Neal

Michelle is a business analyst and strategic project leader with over ten years of leadership experience across multiple industries and seven years within the technology sector. She specializes in workforce analytics, operational performance, and high-volume labor environments, bringing a practical understanding of how data informs staffing strategy, cost control, and organizational efficiency.

Throughout her career, Michelle has led cross-functional initiatives, managed complex projects, and supported teams in fast-paced, high-demand environments. While continuing to expand her technical analytics skillset, she excels at translating data into strategic insights that support executive decision-making and long-term business planning. Her core strengths include strategic thinking, workforce risk assessment and analysis, project leadership, and connecting analytics to measurable business impact.

Skills Considerations

While the team brings strong analytical, strategic, and operational strengths, several areas present potential skill gaps that will be intentionally addressed throughout the project.

1. Limited Coding Experience

The team has minimal traditional coding experience, which may present challenges in areas such as data transformation, modeling automation, and advanced analytical workflows. However, this gap will be mitigated through the use of AI-enhanced tools, low-code analytics platforms, and assisted code generation that reduces the need for manual scripting. These technologies will allow the team to perform robust analysis, validate outputs, and iterate quickly without deep programming expertise.

2. Developing Financial Acumen

While the team is proficient in strategic and operational analysis, financial modeling and cost-benefit evaluation are not core strengths. To ensure that recommendations are fiscally sound, the team has invested time in learning fundamental business finance concepts—such as cost structures, margin analysis, cash flow basics, and return-on-investment evaluation. This foundational understanding enables the team to provide responsible and informed financial guidance, while more complex financial scenarios may require cross-checking assumptions with available academic or expert resources.

3. Technical Systems Integration Limitations

The implementation of integrated technical systems requires expertise beyond the team's current capabilities. While the team can design analytics, define data needs, and build models once systems are in place, the actual configuration and integration of those systems (e.g., POS integration, inventory syncs, financial connectors) will require guidance from technical and IT specialists. The team will rely on experts for proper setup and system alignment to ensure data accuracy and reliability, and will then take ownership of the analytics, insights, and decision-support layers that follow.

Strategic Analysis

External Analysis

The U.S. ice cream and frozen dessert market is a mature but stable industry with consistent consumer demand. Industry estimates place the market value at roughly $14 billion, with expected growth of about 3–4% annually (IBISWorld, 2024; Mintel, 2024). Ice cream remains one of the most widely consumed dessert products in the United States, with most consumers purchasing frozen desserts regularly (Statista, 2025).

While demand remains steady, profitability for small independent ice cream shops can be sensitive to operating costs. Labor is typically one of the largest expenses in food service, often accounting for

30–35% of total operating costs (Bureau of Labor Statistics, 2025). Businesses that experience fluctuating demand throughout the year must manage these costs carefully to remain profitable.

Product costs have also increased in recent years. The average retail price of ice cream reached approximately $6.27 per half gallon in 2024 (Bureau of Labor Statistics, 2025). As ingredient and supply costs rise, smaller food businesses must balance pricing, production volume, and product waste to maintain healthy margins.

Regional Market Environment: Oceanside, New York

Crafted Cones operates in Oceanside, New York, a Long Island community located in Nassau County within the greater New York metropolitan area. The area has a strong residential base while also experiencing increased activity during warmer months due to nearby beaches, parks, and waterfront attractions along the South Shore.

Warmer weather typically drives higher demand for frozen desserts, particularly in communities where outdoor recreation and beach activity are common. During the summer months, increased foot traffic and recreational activity can lead to higher transaction volumes for businesses such as ice cream shops. In contrast, colder winter months generally bring lower demand, creating revenue fluctuations throughout the year.

Oceanside also benefits from its proximity to New York City and surrounding commuter routes, contributing to a mix of local residents, commuters, and visitors within the customer base. For small businesses, this combination can create opportunities for steady local traffic while still benefiting from seasonal increases in demand.

Labor Market Conditions

Labor conditions are another important factor shaping the operating environment for small food service businesses. The restaurant and food service industry continues to face workforce shortages and increasing wage expectations. According to the Bureau of Labor Statistics, fast food and counter-service workers in the United States earned a median hourly wage of approximately $14.65 in 2024 (Bureau of Labor Statistics, 2025).

In regions near major metropolitan areas such as New York City, businesses often compete with larger restaurants, retail stores, and national chains for hourly employees. This can make hiring and retaining staff more challenging for smaller businesses. As a result, efficient scheduling, effective training, and strong employee retention practices become especially important for maintaining service quality and controlling labor costs.

Implications for Crafted Cones

These external factors create both opportunities and challenges for Crafted Cones. Strong summer demand and a dense residential market provide opportunities for growth, but seasonal demand fluctuations, rising labor costs, and competitive pressures can make operations more complex. These conditions will prove to be even more important to consider in the future as Crafted Cones evaluates any expansion opportunities. Understanding these external influences provides important context for the operational and analytical findings discussed in the following sections.

Internal Analysis

Crafted Cones' internal environment reflects a combination of strong brand assets, operational challenges, and strategic growing pains that shape its ability to scale effectively. The business benefits from deep community loyalty and a differentiated local identity, yet faces structural constraints tied to manual operations, inefficient production patterns, and limited data-driven decision-making.

Strengths

Deep Community Connection and Authentic Local Identity

Crafted Cones maintains strong relationships with the local coastal community and differentiates itself through locally inspired flavors and authentic engagement. This connection creates strong emotional loyalty and strengthens the brand's position against larger competitors.

Strong Customer Loyalty and Repeat Visits

The brand's community alignment and personalized customer experience foster repeat visits, providing a dependable customer base even during seasonal fluctuations. This loyalty is a key internal advantage that supports revenue stability.

Agile Small-Batch Production Model

Small-batch production enables Crafted Cones to adapt quickly to changing customer preferences, local trends, and seasonal demand. This operational agility allows the business to experiment with new flavors and respond quickly to evolving customer interests without the constraints of large-scale production systems.

Leadership Openness to System Modernization and Data Integration

Leadership recognizes the limitations of manual processes and is open to adopting integrated CRM or ERP systems that would centralize sales, inventory, labor, and financial data. This forward-thinking mindset positions the business for long-term scalability and more data-informed decision-making.

Operational Challenges and Areas for Improvement

Heavy Reliance on Manual Processes and Lack of a Centralized Operational System

Crafted Cones currently relies on handwritten logs, spreadsheets, and informal processes to track sales, inventory, staffing, and customer behavior. Without a centralized CRM or ERP system to unify operational data, leadership must manually reconcile information across multiple sources. This fragmented structure limits real-time visibility, slows decision-making, and restricts the business's ability to anticipate seasonal demand or respond quickly to operational changes.

Limited Data Integration for Strategic Decision-Making

Without unified operational data, leadership faces challenges when forecasting demand, planning ingredient purchasing, or optimizing staffing during peak and off-peak periods. These limitations are especially significant given the highly seasonal nature of the business.

Static Reporting Limits Operational Agility

With no automated dashboards or real-time performance monitoring, reporting remains static and backward-looking. This makes it difficult for leadership to adjust quickly to variables such as weather patterns, school schedules, or tourism surges, all of which significantly influence customer traffic.

Ineffective Off-Season Promotions Reduce Profitability

During off-peak months, Crafted Cones frequently relies on promotions that erode margins without delivering meaningful customer retention improvements. These promotions treat all customers similarly, failing to distinguish between local residents with long-term value and seasonal visitors with shorter engagement cycles. Industry research shows that restaurant operators are increasingly using data and analytics to target offers more precisely, improving profitability and customer engagement compared to broad, undifferentiated promotions (McKinsey & Company, 2023).

Flavor Production Inefficiencies and Waste

Top-selling flavors generate the majority of sales, yet niche and seasonal flavors—while important for brand identity—introduce variability in demand that requires careful production planning. Industry research indicates that consumers enjoy both familiar flavors and new options, with many viewing the opportunity to try unique flavors as part of the overall ice cream experience (Mintel, 2024). Excess production not only impacts profitability but also complicates inventory planning and labor scheduling.

Analytics Opportunity

Our team identified the analytics opportunity by conducting a detailed review of Crafted Cones' current operational practices, data collection methods, seasonal demand patterns, and decision-making

processes. Through examining the company's reliance on manual spreadsheets, handwritten logs, and informal reporting workflows, it became clear that the business lacks a centralized data environment capable of supporting informed, proactive management. This discovery aligns with leadership's recognition that operational blind spots, inconsistent visibility, and the absence of integrated systems limit the business's ability to plan effectively, manage inventory, forecast demand, and optimize labor.

As we mapped how sales data, staffing logs, ingredient usage, production plans, and customer interactions were manually tracked across disconnected sources, the analytics opportunity became evident: Crafted Cones requires a unified system foundation paired with a structured analytics roadmap to transition from reactive decision-making to predictive, data-informed management. This opportunity is best captured through a three-phase approach that introduces a scalable path for system integration, insight development, and long-term strategic improvement.

Phase 1: Establish a Unified CRM/ERP System Foundation

The absence of a centralized operational system is the primary barrier limiting visibility, accuracy, and efficiency across the business. Implementing an integrated CRM and ERP framework would consolidate essential operational data—including sales, inventory, production, workforce scheduling, customer interactions, and vendor management—into a single environment. This foundational step replaces fragmented information with consistent, reliable data, creating the conditions necessary for more advanced analytics across the organization.

Phase 2: Build Analytics Capabilities Across Three Strategic Pillars

With centralized data in place, Crafted Cones can transform raw operational information into actionable insights that answer key business questions.

Operational Insights

Analytics can reveal daily and seasonal sales patterns, optimize production schedules, reduce ingredient waste, and highlight how external variables—such as weather, tourism, and school calendars—affect demand. These insights are particularly valuable for addressing inefficiencies in flavor production and managing inconsistent demand throughout the year.

Workforce Insights

Digitized labor data enables analytical models that identify peak staffing periods, align schedules with forecasted sales, reduce overscheduling, and improve employee productivity. Workforce analytics are especially important for navigating the seasonal hiring and labor fluctuations that characterize the business.

Market & Customer Insights

A CRM system enables meaningful customer segmentation, allowing leadership to distinguish between local residents and seasonal visitors—something that is not possible with current manual records. Analytics can reveal visit frequency, flavor preferences, customer lifetime value, and the effectiveness of promotional campaigns. These insights directly address existing challenges associated with broad, undifferentiated promotional strategies.

Phase 3: Apply Insights to Strengthen Operations and Guide Strategic Growth

Once insights are generated, the business can apply them to optimize operations and refine customer strategy. Forecast-driven production planning can reduce product waste, labor scheduling can become more precise, and customer engagement strategies can shift from broad promotions toward targeted, data-driven outreach.

Strategic Impact

The analytics opportunity for Crafted Cones lies in transforming a manually operated business into a data-enabled organization through a clear three-phase roadmap. By establishing a unified system foundation and building analytics capabilities across operational, workforce, and market and customer domains, Crafted Cones can improve efficiency, reduce waste, strengthen customer relationships, and pursue strategic growth with greater confidence and precision.

Rationale

Crafted Cones has operational constraints that stem from fragmented data sources, manual reporting processes, and limited visibility into sales, workforce patterns, inventory movement, and customer behavior. These gaps restrict the company's ability to anticipate seasonal demand, optimize labor and production, manage inventory efficiently, and tailor customer engagement strategies. With decisions currently driven by intuition, experience, and manual spreadsheets, the business lacks the real-time insight required to operate proactively in a highly seasonal, tourism-influenced market.

Analytics provides a structured and scalable solution to these challenges by enabling the business to replace reactive decision-making with predictive, data-driven planning. Introducing integrated systems such as a CRM or ERP platform will unify sales, inventory, labor, and customer data into a single source of truth. This centralized foundation is essential for producing accurate insights, reducing operational waste, improving demand forecasting, and strengthening financial performance. It directly addresses inefficiencies in flavor production, staffing variability, and broad, undifferentiated promotional strategies.

The three-phase approach—establishing a unified system foundation, developing analytics capabilities, and applying insights to improve strategy—ensures that Crafted Cones can adopt analytics at a manageable pace while achieving both immediate and long-term business impact. This structured model

reduces implementation risk, increases organizational readiness, and supports scalability, particularly as the company evaluates flexible expansion options such as a mobile ice cream truck.

Ultimately, analytics is the right solution because it aligns with leadership's goals, strengthens operational consistency, and transforms previously disconnected data into actionable insight. By embedding analytics across operational, workforce, market and customer, and expansion decisions, Crafted Cones will gain the clarity and confidence needed to improve performance, reduce waste, and pursue growth opportunities that reflect the unique dynamics of its seasonal business.

Stakeholder and Requirement Analysis

Crafted Cones' success depends on a diverse group of stakeholders whose needs, challenges, and expectations shape the requirements of the analytics solution. The absence of integrated systems, combined with heavy reliance on manual workflows, impacts each stakeholder group differently and highlights the importance of implementing a centralized CRM/ERP foundation to improve visibility, coordination, and decision-making across the business.

Owners

Role: The owners are responsible for strategic planning, financial oversight, and long-term growth decisions.

Current Challenges:

- Difficulty understanding the financial impact of extreme seasonality and tourism patterns
- Limited visibility into real-time operational metrics, including labor costs, waste patterns, and flavor performance
- Concerns about rising expenses tied to manual scheduling, overproduction, and broad discounting that erodes profit margins

Requirements:

- Unified dashboards that summarize operational, workforce, and customer performance indicators.
- Predictive tools to forecast demand based on weather, tourism, and seasonality.
- Clear financial insights that support long-term strategic planning and growth decisions

Store Manager

Role: The store manager oversees all daily operations, including scheduling, production planning, inventory management, quality control, and customer service.

Current Challenges:

- Manual scheduling without demand forecasts, leading to understaffing during peak periods or overscheduling during slower periods
- Managing a workforce with varying training levels and seasonal turnover
- Addressing customer complaints driven by long lines, slow service, or inventory shortages
- Forecasting inventory needs and flavor production without access to historical analytics or real-time data
- Reliance on handwritten logs and informal processes that limit operational visibility

Requirements:

- Forecast-driven scheduling tools with automated staffing recommendations.
- A labor dashboard displaying hours, overtime, productivity, and training status.
- Alerts for waste thresholds, low inventory, or flavor stockouts.
- Tools that reduce administrative burden and increase accuracy through automated reporting and centralized data.

Employees

Role: Full-time and part-time staff are responsible for customer service, register operations, product preparation, and maintaining store cleanliness.
Current Challenges:

- Uneven staffing patterns that cause burnout during peak traffic periods
- Lack of consistent and accessible training, particularly for part-time or seasonal employees
- Limited transparency around scheduling expectations or schedule changes
- Feeling unsupported during peak seasons due to overstretched management resources

Requirements:

- Fair, predictable scheduling informed by accurate demand forecasting.
- A clear and standardized onboarding/training framework.
- Defined shift leader or “captain” roles to reduce pressure during high-volume times.
- Recognition and engagement programs (e.g., “Captain Cone” employee of the month).

Customers

Role: Customers include local residents, repeat visitors, and tourists who expect high-quality products and an efficient service experience.

Current Challenges:

- Long wait times during peak weekends or tourist surges.
- Irregular availability of top flavors due to inefficient production planning.
- Limited awareness of new products, upcoming events, or rotating flavors.

- Frustration stemming from staff shortages or slow service.

Requirements:

- Consistent availability of high-demand flavors supported by improved production forecasting
- Reduced wait times through optimized staffing and operational planning
  Enhanced digital engagement, such as flavor voting or event notifications
- Expanded opportunities to interact with the brand, including service through a future mobile ice cream truck

Success Criteria

The success of the proposed analytics solution will be evaluated through a combination of system adoption metrics, operational performance improvements, and measurable business outcomes. These criteria ensure that the implementation of centralized systems, analytics dashboards, and forecasting tools produces meaningful improvements in decision-making, operational efficiency, and customer engagement.

Within the first 90 days of implementation, 100% of employees will be onboarded and trained on the new system, with consistent use of the platform integrated into daily operational workflows. This includes the use of centralized systems for capturing sales data, tracking inventory, managing workforce schedules, and recording customer interactions. Successful adoption ensures that the organization moves away from fragmented manual processes toward a structured, data-driven operating environment.

In addition to system adoption, the project establishes key performance indicators (KPIs) across the three primary analytics pillars: Operational, Workforce, and Market & Customer analytics. At least two KPIs will be defined for each pillar and monitored through centralized dashboards on a quarterly basis. Final KPI targets will be established in collaboration with company leadership to ensure they reflect realistic operational goals and strategic priorities (Quantic School of Business and Technology, 2025, p. 179).

Operational Performance

Operational KPIs will focus on improving production efficiency, reducing waste, and strengthening inventory management through data-driven planning. Key indicators may include improved ingredient yield, reduced product waste, and increased inventory turnover for perishable ingredients. Forecast accuracy will also be measured by comparing projected demand with actual sales data, allowing the business to refine production planning and reduce overproduction during slower periods.

Workforce Performance

Workforce KPIs will evaluate improvements in labor efficiency, employee engagement, and training consistency. Key indicators may include improvements in employee retention, increased completion of required training programs, and stronger alignment between scheduled labor hours and actual customer demand. These metrics support more effective scheduling decisions and reduce operational strain during peak seasonal periods.

Market & Customer Performance

Market and customer KPIs will measure improvements in customer engagement and the effectiveness of marketing initiatives. Indicators may include reductions in customer acquisition cost, improvements in customer sentiment and satisfaction, and increases in repeat visits from local customers. With the implementation of a CRM system, the business will also gain greater visibility into customer preferences, visit frequency, and promotional performance.

Forecasting and Financial Outcomes

Forecasting models will be used to project sales trends, inventory requirements, and seasonal demand patterns. Success will be measured by comparing forecasted values against actual outcomes and using these insights to refine production, staffing, and purchasing decisions over time.

By combining improved operational planning, workforce efficiency, and targeted customer engagement strategies, the analytics initiative aims to support measurable financial improvements. The long-term objective is to reduce operational inefficiencies while increasing revenue opportunities, with a target of achieving up to a 20% improvement in overall sales performance within one year of implementation.

Scope and Schedule of Deliverables

Feasibility

The feasibility assessment for this project focused on evaluating the technical, operational, and economic requirements of implementing an integrated CRM/ERP system and accompanying analytics framework for Crafted Cones.

Initial review indicates that the proposed solution is feasible for several reasons. First, the business currently operates using fragmented manual workflows across sales, inventory, labor scheduling, and customer interactions. While these processes are not centralized, they follow consistent patterns that can be structured into a unified data environment. Second, the organization already collects key operational data in manual formats such as spreadsheets and handwritten logs. Although migrating this information will require effort, it provides a foundation for structured data capture moving forward.

Operational feasibility is also strong due to leadership support and clear recognition of the limitations of current processes. The organization has expressed a clear need for improved forecasting, reporting, and operational visibility. Finally, economic feasibility appears favorable. Expected benefits—including reduced ingredient waste, improved labor scheduling, and more targeted customer engagement—are projected to outweigh the costs associated with implementing centralized systems and analytics tools.

Overall, the feasibility assessment supports proceeding with the proposed analytics roadmap and CRM/ERP system framework as a scalable modernization effort that can improve operational efficiency and strategic decision-making.

Scope

The scope of this project is to guide Crafted Cones through the transition from manual, non-integrated operations toward a scalable, analytics-enabled operating model. This work follows the three-phase CRM/ERP and analytics maturity framework outlined earlier in the project.

In Scope

1. CRM/ERP System Blueprint and Data Foundation Assessment

This component focuses on documenting current workflows and identifying the data structures required to support a centralized operational system.

Activities include:

- Mapping existing manual processes and data sources, including sales tracking, inventory logs, labor schedules, and customer feedback
- Identifying integration requirements between potential CRM, POS, inventory, and workforce systems
- Recommending data structures necessary to support a unified CRM/ERP environment

2. Analytics Framework Across Three Strategic Pillars

Analytics development will focus on the three pillars central to Crafted Cones' operational strategy.

Operational Analytics

- Demand forecasting
- Production planning optimization
- Ingredient waste reduction

Workforce Analytics

- Predictive scheduling
- Labor productivity insights
- Training and workforce performance tracking

Market & Customer Analytics

- Customer segmentation
- Promotional campaign effectiveness
- Flavor preference analysis and demand trends

3. Prototype Dashboards Using Simulated Data

Until the CRM/ERP system is implemented and data is fully and routinely captured, prototype dashboards will be developed using representative simulated datasets to demonstrate how integrated CRM/ERP data could support decision-making.

These dashboards will include:

- Operational performance dashboards
- Workforce management dashboards
- Market and customer analytics dashboards

The prototypes will be built using Power BI to illustrate how data visualization tools can transform operational data into accessible business insights.

4. Insights and Strategic Recommendations

Only after a full business cycle of data is available in the CRM/ERP system (roughly 12-18 months post implementation and training), and based on the analysis and dashboard outputs, the project will produce strategic recommendations designed to improve operational performance and support future growth.

Recommendations may include:

- Staffing optimization strategies
- Production planning improvements
- Flavor portfolio and product mix analysis
- Targeted promotional strategies

Out of Scope

The following activities fall outside the scope of this project:

- Full technical implementation of a CRM/ERP system

- Integration of live POS or financial data into a production system
- Deployment of fully automated predictive models beyond analytical prototypes

These elements would typically occur during a future operational implementation phase following the strategic analysis and planning completed in this project.

Schedule of Deliverables

The project schedule follows a phased structure aligned with the analytics roadmap described earlier in the report. Each phase builds progressively toward a more advanced analytics capability for the organization.

Phase 1 – Foundation and Business Understanding

Deliverables

- Documentation of current operational workflows and data collection processes
- Assessment of the current data landscape and data quality considerations
- Identification and definition of key performance indicators (KPIs) across the three analytics pillars

Phase 2 – Prototype Analytics and Dashboard Development

Deliverables

- Development of prototype forecasting structures using simulated CRM/ERP data
- Power BI dashboard prototype (Operations, Workforce, Marketing, and Customer analytics modules)
- Preliminary analytical insights related to key operational and workforce performance indicators

Phase 3 – Insights and Strategic Roadmap

Deliverables

- Comprehensive insight analysis across the three analytics pillars
- Operational and workforce optimization recommendations
- CRM/ERP and analytics implementation roadmap to support long-term system integration and strategic growth

Data Sources and Structure

The analytics environment for Crafted Cones uses a multi-year Excel dataset designed to simulate the company's transition from manual recordkeeping to a more structured CRM and ERP-style data environment. The dataset aligns with the project narrative, which describes how the business initially relied on handwritten logs and fragmented spreadsheets before moving toward centralized systems capable of supporting forecasting, reporting, and data-driven planning.

All years are stored in the same Excel tables so Power BI can ingest the data as a unified source. A Year field identifies whether each record comes from the manual phase or the future structured phase. Year 1 contains migrated manual entries, while Years 2 and beyond include fictional but realistic records that represent the type of information a CRM and ERP system would produce.

Each table in the Excel workbook follows a consistent format across years, which allows Power BI to combine and model the data efficiently. The shared **Year** field makes it possible to analyze operational differences over time while maintaining a single table structure. The dataset captures information across three primary areas of the business: **operational data, workforce data, and market & customer data**. This design preserves the authenticity of the Year 1 manual data while also demonstrating the benefits of structured **CRM/ERP-style** data in later years.

These three data categories collectively support the analysis of Crafted Cones' operational performance, workforce management, and customer engagement. The complete relational schema for the dataset is provided in the Appendix, and the full data dictionary is included in the Appendix. Across the three data categories, the dataset contains **11 tables** with varying data types. For clarity, Table 1 summarizes the number of tables within each business function. The **Market and Customer** category contains the largest number of tables and includes a mix of character, date, and integer data types. Collecting both qualitative and quantitative data in this way enables the owners of Crafted Cones to make more informed decisions that incorporate both operational performance and customer perspectives.

Business Function	Operational	Workforce	Market and Customer
# Tables	2	4	5

Table 1: Number of data tables per business function

Gathering data from multiple areas will allow the owners of Crafted Cones to make nuanced yet strategic decisions about the future of their business. Each area of data will be further explored below.

Operational Data

The operational data consists of two tables as demonstrated in table 1. Table 2 demonstrates the main goal of each of these tables. In summary, the operational tables aim to provide insight into multiple aspects of the business' operations, such as the financials and product movement.

	Data Type	Table Name	Summary of Data Table
1	Operational	Product_Details	Table outlines product details available at Crafted Cones.
2	Operational	Sales_Transactions	Table outlines sales transactions at Crafted Cones.

Table 2: Crafted Cones operational data tables.

Figure 1 demonstrates the relational schema for the operational data. The Product_Details table has Product_ID as the primary key, this is also an alphanumeric value specifically relating to a product offering. Product_ID is a foreign key in the Sales Transaction table. This allows more expansive data analysis through the operational data, accessing attributes of both tables.

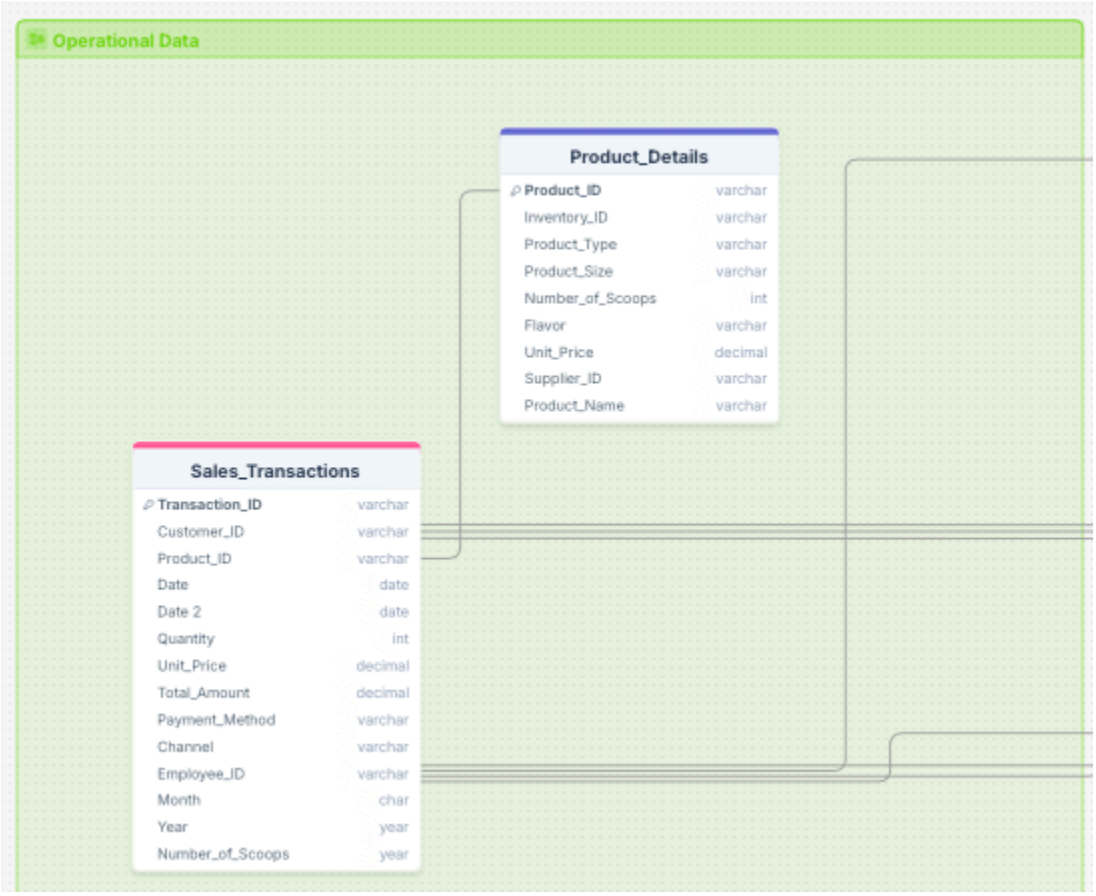

Figure 1: Crafted Cones operational data in the relational schema

Workforce Data

The workforce data consists of four tables as demonstrated in table 1. Table 3 demonstrates the main goal of each of these tables. In summary, the workforce tables aim to provide insight into multiple aspects of the workforce, such as employee demographic and administrative data, training status, as well as allowing Crafted Cones to maintain accurate records concerning the departure of employees.

	Data Type	Table Name	Summary of Data Table
1	Employee	Employee_Details	The table outlines various demographic and administrative aspects of Crafted Cones' employees.
2	Employee	Employee_Shift	The table outlines employee shift information.
3	Employee	Employee_Training	The table outlines the completion status of employee training.
4	Employee	Employee_Turnover	The table outlines employee departures and submissions to an employee exit survey.

Table 3: Crafted Cones workforce data tables.

Figure 2 demonstrates the relational schema for the workforce data. The primary key in two of the tables (Employee_Details and Employee_Turnover) is Employee_ID. Each employee is provided an alphanumeric code which associates them to their administrative and training information. Employee_ID acts as a foreign key in the other two tables (Employee_Shift and Employee_Training), facilitating extensive workforce data analytics allowing calculations to be undertaken using all four tables. Employees who are also supervisors will also have their Employee_ID appear under Supervisor_ID.

Training_ID is the primary key in the Employee_Training table. This table outlines the completion status of each employee's mandatory and elective training. Training_ID is alpha numeric, and assigned to each training activity.

There are two distinct shifts at Crafted Cones. Shift_ID is the primary key in the Employee_Shift table. Each of the two shifts are assigned a specific alpha numeric code. This will allow for more reliable data analysis and interpretation as the owners of Crafted Cones will not only have to rely on date specific analysis.

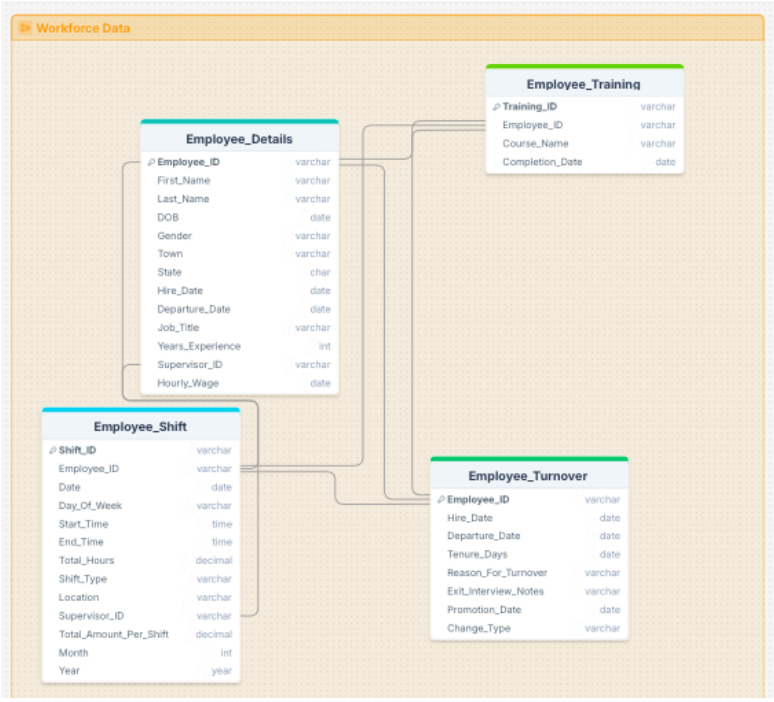

Figure 2: Crafted Cones workforce data in the relational schema

Market and Customer Data

The market and Customer data consists of five tables as demonstrated in table 1. Table 4 demonstrates the main goal of each of these tables. In summary, the market and customer tables aim to provide insight into multiple aspects of the marketing and customer interactions with Crafted Cones, such as customer demographic and administrative data, customer satisfaction survey responses, administrative details of marketing campaigns, as well as Crafted Cones' social media usage.

	Data Type	Table Name	Summary of Data Table
1	Marketing	Marketing_Social	The table outlines administrative information of marketing campaigns on social media.
2	Marketing	Marketing_Tactics	The table outlines marketing tactics used for promotional campaigns.
3	Customer	Customer_Personal	The table outlines various demographic and administrative aspects of Crafted Cones customers.
4	Customer	Customer_Preference	The table outlines the communication preferences and discount sensitivity of Crafted Cones customers.
5	Customer	Customer_Satisfaction	The table outlines responses by customers to satisfaction surveys.

Table 4: Crafted Cones customer and marketing data tables.

Figure 3 demonstrates the relational schema for the market and customer data. The market and customer data and be further divided into marketing data (2 data tables) and customer data (3 data tables). Within the market data, the primary key in both tables (Marketing_Social and Marketing_Tactics) is Campaign_ID. This variable refers to a specific alpha numeric identification code assigned to each marketing campaign undertaken in any format (e.g. print, digital, etc.).

Crafted Cones could gain access to customer data through the creation of a digital customer loyalty program. Customer_ID is an automatically generated alphanumeric code generated upon enrollment in the program. Each new customer account creation will be assigned a Customer_ID. The Customer_Personal and Customer_Preference tables utilize Customer_ID as the primary key. The Customer_Satisfaction table uses Customer_ID as a foreign key. Survey_ID is the primary key in the Customer_Satisfaction table. After completing a purchase at Crafted Cones, a customer survey is sent via the email provided at the time of loyalty enrollment. This survey_ID is a unique alphanumeric identification code.

Figure 3: Crafted Cones workforce data in the relational schema

Exploratory Data Analysis

All visualizations, datasets, and supporting documentation for this project are available in the Appendix and in the consultancy's GitHub repository.

Crafted Cones historically relied on manual recordkeeping; artificial intelligence was used to supplement portions of the dataset in order to create a proof-of-concept analytics environment. The prompts used to generate the simulated data are included in the Appendix[2]. This approach allowed the dataset to simulate the type of structured information that could be exported from a CRM or ERP system as proposed in Phase 1 of the analytics roadmap.

Phase 2 of the analytics plan focuses on transforming centralized data into insights that answer core business questions and guide decision-making. To support this objective, the analytics team developed an interactive dashboard environment and conducted exploratory analysis across operational, workforce, and customer data. In addition, preliminary predictive modelling was explored to evaluate how forecasting could support staffing decisions and seasonal planning.

The relevant exploratory analysis methods are discussed in the following subsections.

[2] Consultant team note: Per Quantic's *Use of AI Policy*, this report uses fictitious data to demonstrate the potential deliverables to the client in lieu of phase 1 of our proposed data plan being completed (Adoption of a CRM system). Any business analytics reporting provided to a real client would not include fictitious data.

Dashboard Predictive Analysis

The predictive analysis undertaken to be included in the dashboard had the following steps undertaken. Manually conducted sales data from 2023 was utilized for this analysis, along with data supplemented by artificial intelligence (AI) for demonstrative purposes[3]. Crafted Cones will be advised to not make business decisions using the predictive modelling utilizing AI data.

1. Data Cleaning and Normalization

    a. The data cleaning stage included validating data formatting, types as well as ensuring the entire dataset was uploaded into Python. After initial cleaning was undertaken, the data cleaning stage was validated to confirm all actions were completed as intended. Table 5 and table 6 summarize the results were achieved as expected.

Dtypes (sample):	
Transaction_ID	object
Customer_ID	object
Product_ID	object
Date	object
Date 2	object
Quantity	int64
Unit_Price	object
Total_Amount	object
Payment_Method	object
Channel	object
Month	int64
Year	int64
Number_of_Scoops	int64
Total_Amount_raw	object
Total_Amount_clean	float64

[3] Consultant team note: Per Quantic's *Use of AI Policy*, this report uses fictitious data to demonstrate the potential deliverables to the client in lieu of phase 1 of our proposed data plan being completed (Adoption of a CRM system). Any business analytics reporting provided to a real client would not include fictitious data. At the time of project proposal the consultancy team used GitHub's Copilot artificial intelligence to create python code.

Dtypes (sample):	
Unit_Price_raw	object
Unit_Price_clean	float64
Quantity_clean	int64
Number_of_Scoops_clean	int64
Year_clean	int64
Month_clean	int64

Table 5: Data types in dataset after data cleaning

Quick summary for cleaned columns (sample):			
	Count	Mean	Std
Total_Amount_clean	1000.0	11.7340	5.288167
Unit_Price_clean	1000.0	5.7315	1.150209
Quantity_clean	1000.0	2.0480	0.816310
Number_of_Scoops_clean	1000.0	2.4960	1.114103
Year_clean	1000.0	2023.0000	0.000000
Month_clean	1000.0	1.0000	0.000000

Table 6: Summary of cleaned columns

2. Exploratory Data Analysis: Outliers, descriptive statistics, data visualizations, customer behavior analysis

    a. First, calculation of missing data was undertaken confirming there was no missing data other than within the “Channel” variable, shown in table 7. This wasn't initially anticipated because all rows were created with a channel via artificial intelligence. The data was maintained as is, as this was not a needed variable for the purpose of undertaking analysis.

Dtypes (sample):	
Transaction_ID	0

Customer_ID	0
Product_ID	0
Date	0
Date 2	0
Quantity	0
Unit_Price	0
Total_Amount	0
Payment_Method	0
Channel	7586
Month	0
Year	0
Number_of_Scoops	0
Total_Amount_raw	0
Total_Amount_clean	0
Unit_Price_raw	0
Unit_Price_clean	0
Quantity_clean	0
Number_of_Scoops_clean	0
Year_clean	0
Month_clean	0

Table 7: Summary table of missing data

b. To further explore the data the Interquartile range (IQR) and z-scores were calculated. This was used to identify what entries would be considered outliers in the data and further understand the distribution of the dataset. Outlier analysis was then placed in visualizations utilizing boxplots and histograms to visualize the distribution of data.

c. Descriptive statistics were next completed to understand the characteristics of the dataset. Example frequency is noted in table 9 below indicating transactions per month in 2023.

--- Month and Year Frequency ---				
Month	Month_clean	Year_clean	Count	Percentage
0	1	2023	1913	5.09
1	2	2023	1800	4.79
2	3	2023	1929	5.13
3	4	2023	1764	4.69
4	5	2023	1671	4.44
5	6	2023	6890	18.32
6	7	2023	7384	19.63
7	8	2023	6579	17.49
8	9	2023	1821	4.84
9	10	2023	1810	4.81
10	11	2023	2013	5.35
11	12	2023	2042	5.43

Table 8: Frequency of transactions per month in 2023

d. Customer behavior analysis was next conducted once it was confirmed the data had no existing data quality concerns. Multiple analyses were undertaken to understand the behaviour of consumers. One example is outlined in figure 4 below noting the average transaction value per month.

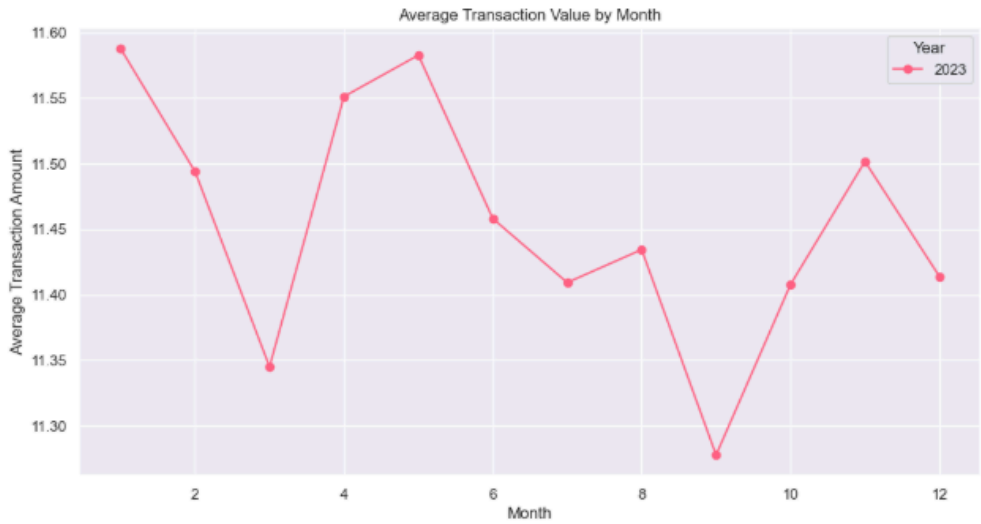

Figure 4: Example average customer transactions per month

3. Sales Forecasting: Baseline Modelling

Next baseline models are established. The aim of this step is to utilize a baseline model to further understand the current reality of the business' sales. First the data from the last 3 months is reserved to test the predictions of the model, this is referred to as "test/train split" (refer to GitHub repository for output). Next, three forecast models were utilized:

- **Naive:** Assume next month's sales are equal to last month's sales.
- **Seasonal Naive:** Uses the same month from the previous year.
- **Moving Average (MA3):** Uses the average of the last 3 months.

The results of the three baseline models are presented below, but first must be validated.

a. Validate model using rolling-origin backtesting

The results of the three baseline models are validated using rolling-origin backtesting with examples of the output showin figure 5 and table 9 below. This method of validation is more robust that the test/train split and accounts more aptly for seasonality (refer to GitHub repository for output). Rolling-orgin backtesting starts with a minimum dataset and forecasts the next month's sales, it then continues forecasting one month at a time until all periods have been tested (refer to GitHub repository for output).

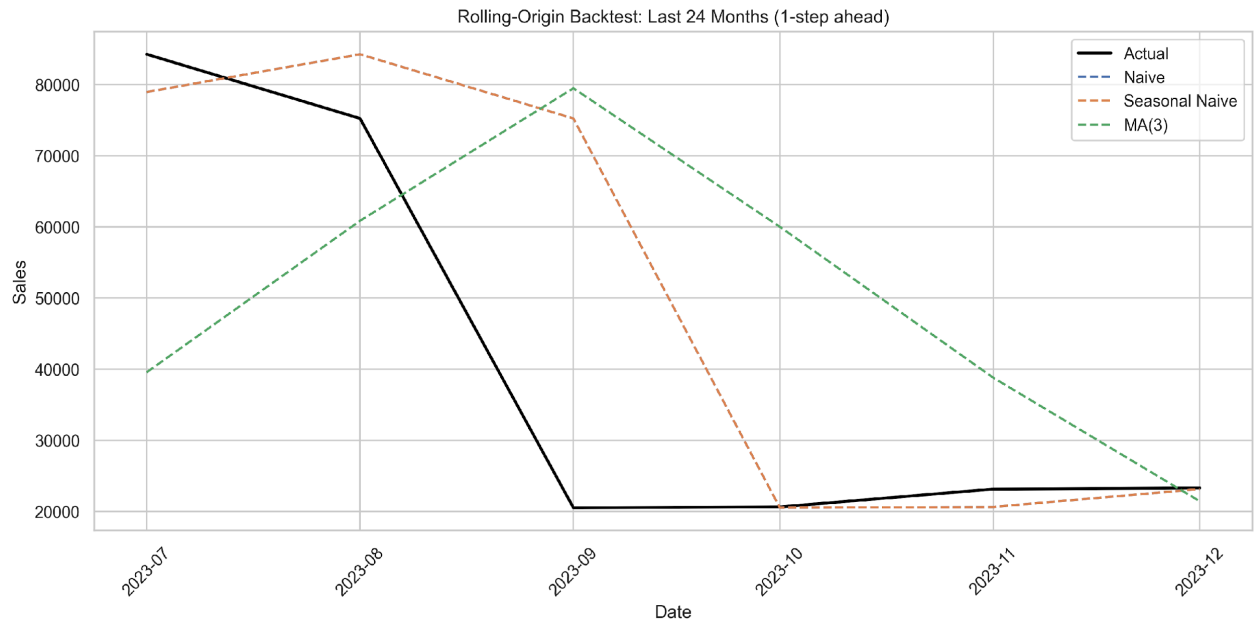

Figure 5: Previous 24-Months Rolling-Origin Backtest Graph

	MAE	RMSE	MAPE%	sMAPE%
Naive	11963.333	22755.18	49.435	24.112
Seasonal naive	11963.333	22755.18	49.435	24.112
MA3	29144.806	35294.935	104.221	61.27

Table 9: Example outputs from time series forecasting models tested

The results demonstrate that there is a strong seasonal component. Both the naive and seasonal naive performed equally well (MAE & RMSE). For both of these baseline forecasts, they are within 24% of actual sales (sMAPE%). The MA3 findings are more pronounced, suggesting baseline forecasts are within 61% of actual sales.

4. Sales Forecasting: Advanced Forecasting Models (SARIMA and Prophet)

Advanced sales forecasting models allow for better accuracy. Running a SARIMA (Seasonal AutoRegressive Integrated Moving Average) test will allow better trends to be captured and account for seasonality (refer to GitHub repository for output). This method uses past sales, and forecasting errors, to predict future sales values (refer to GitHub repository for output). Alternatively, the Prophet test is designed for business operations to separate trend, seasonality, and holiday effects (refer to GitHub repository for output). Prophet is good to be used in this case, as it allows for data with seasonality components (refer to GitHub repository for output).

a. Validate using rolling-origin backtesting

As demonstrated in figure 6 and table 10 below, the findings of the validation of these two models is noted below.

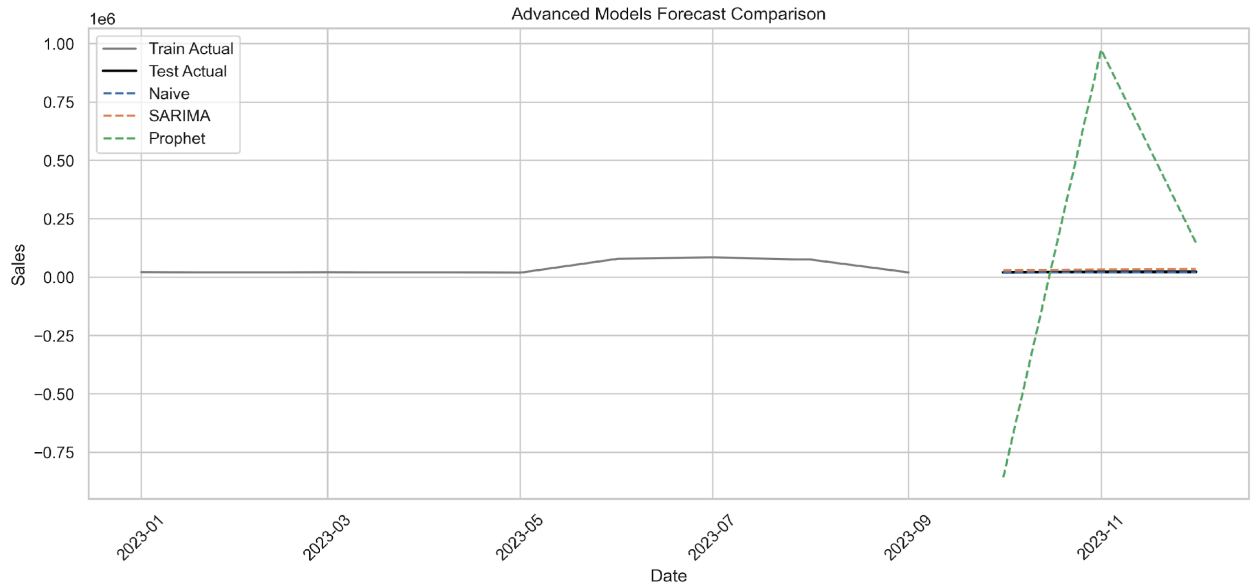

Figure 6: Graph of advanced forecasting model comparison

	MAE	RMSE	MAPE%	sMAPE%
Naive	1833.0	2201.283	7.91	8.386
SARIMA	10277.263	10362.751	45.79	37.219
Prophet	651451.676	750904.92	2965.932	178.792

Table 10: Table comparing performance of advance forecasting models

The results demonstrate that the naive model is best performing with an MAPE of 7.9%. When utilizing rolling-origin backtesting Prophet modelling was more volatile than SARIMA modelling. The results suggest naive models are more accurate than advanced models (SARIMA, Prophet). Though, it is a known data limitation that there is limited data being utilized by these models at this time. As more data is collected, more advanced models may begin performing better than the naive model.

5. Scenario Testing: Seasonal Pattern Modifications: e.g. Early start to season, Late start of season, dampened seasonality.

   Further testing the model with different scenarios will allow the owners of Crafted Cones to see their sales predictions in four different situations with an example shown in figure 7 (refer to GitHub repository for output):

a. Early Season Start: Peak season begins 1 month earlier than historical patterns
b. Extended Peak Season: Peak season lasts 2 months longer than usual
c. Amplified Seasonality: Seasonal peaks are 50% higher, valleys 25% lower
d. Dampened Seasonality: Reduced seasonal variation (more stable year-round sales)

Running these forecasts in Python, demonstrates that the Naive model performed most accurately and reliably when compared with more advanced models (SARIMA and Prophet). Similarly, the Naive and seasonal naive had a consistency score of 0.94, while the other models had much lower consistency scores suggesting less reliability (Prophet = 0.64, SARIMA – 0.30).

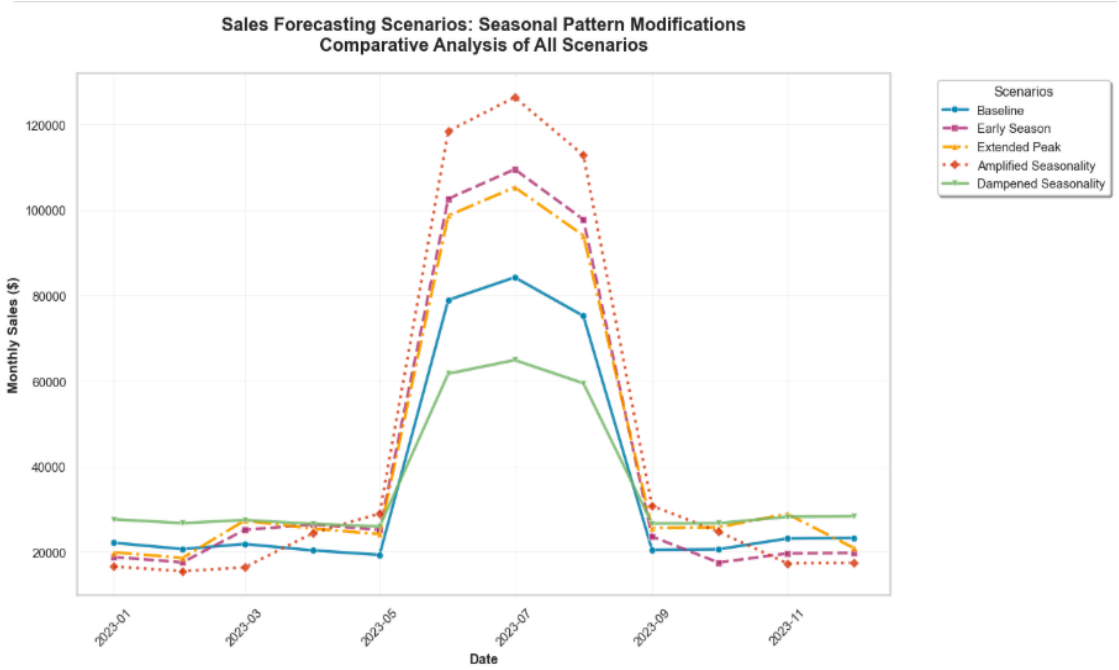

Figure 7: Example graph of seasonal pattern forecasting using simulated data

6. Identifying Business Implications, Method Limitations, and next steps

Finally, when undertaking sales forecasting the business implications and method limitations should be taken into account. The business implications of Crafted Cones' using sales forecasting include:

- Crafted Cones can use the naive model to suggest profits over the next quarter.
- In the case of seasonal changes, the simulated performance (e.g. early start to season) can provide context and strategic insight as to navigating business challenges.
- Time-series forecasting will allow Crafted Cones to strategically plan, and make proactive decisions, for the coming year.

Though sales forecasting would be powerful in its ability to allow for Crafted Cones to strategically plan, the owners should be aware of potential methodological limitations such as the following when interpreting the data:

- Currently, only 12 months of data exists. Once more data is collected, more nuanced results may be demonstrated by forecast modelling.
- Data was previously collected manually which may limit the accuracy of the model.
- Manually collected data is highly prone to human error.
- Sudden or unexpected changes may not be fully captured in modelling.

As outlined previously, these methodological limitations would be in part mitigated if the Customer Relationship Management (CRM) and Enterprise Resource Planning (ERP) system from phase 1 of the data plan is implemented. This will centralize and standardize data collection to the benefit of the analysis and data products discussed.

Methods and Frameworks

Overall Analytical Approach

To guide the analysis for Crafted Cones, this project follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework. CRISP-DM provides a structured and repeatable methodology for analytics projects and ensures that analytical work remains aligned with business objectives throughout the project lifecycle.

The CRISP-DM framework consists of six phases (Quantic, *Solving Problems with Business Analytics*, 2025). For this project, each phase is tailored to the operational realities of Crafted Cones and the structure of the Excel-based dataset that models both the manual Year 1 data and the more structured CRM and ERP-style data from future years.

1. Business Understanding

The first phase defines the analytics goals in the context of the operational challenges described in the Crafted Cones narrative (Quantic, *Solving Problems with Business Analytics*, 2025). The business currently faces issues related to manual processes, strong seasonality, product waste, labor imbalances, limited operational visibility, and the absence of centralized data systems.

The primary objectives of this analytics initiative include improving demand forecasting, optimizing staffing levels, supporting more efficient inventory planning, and generating customer insights.

Key outputs of this phase include:

- Clear problem definition
- Alignment with Crafted Cones' three-phase transformation roadmap

- Identification of analytics priorities across the Operational, Workforce, and Market & Customer pillars

2. Data Understanding

The data understanding phase focuses on exploring and validating the multi-year Excel dataset used in this project (Quantic, *Solving Problems with Business Analytics*, 2025). The dataset contains both manually recorded operational data from Year 1 and simulated CRM/ERP-style records for future years.

Initial analysis examined patterns, missing values, inconsistencies, and seasonal variations within the dataset. Particular attention was given to identifying differences between manually recorded data and the more structured data expected from future systems.

Key outputs of this phase include:

- Initial exploration of sales, labor, inventory, and customer variables
- Identification of potential data quality issues
- Recognition of seasonal demand patterns and operational peaks

3. Data Preparation

Data preparation represents one of the most resource-intensive phases of the analytics process. In this project, preparation involved cleaning manually recorded data, standardizing formats across multiple years, and integrating the dataset into a structured format suitable for analysis (Quantic, *Solving Problems with Business Analytics*, 2025).

Key preparation steps included:

- Cleaning Year 1 manually recorded data
- Standardizing variable formats across all years
- Tagging each observation with a Year identifier
- Structuring the dataset using a star schema architecture in Power BI

The resulting data model includes:

- **Dimension tables**
    - Date
    - Product
    - Employee
    - Customer
    - Weather
- **Fact tables**
    - Sales transactions
    - Labor shifts

- Inventory movement

This structure supports efficient analytics and dashboard development.

4. Modeling

The modeling phase focuses on selecting and applying analytical techniques to answer the business questions identified earlier in the project (Quantic, *Solving Problems with Business Analytics*, 2025). Multiple types of analytics were applied to support Crafted Cones' operational and strategic objectives.

Descriptive modeling

- Trend analysis of sales, waste, labor, and customer behavior
- Visual summaries built in Power BI
- Baseline ratios such as sales per hour, waste percent, flavor contribution

Diagnostic modeling

- Identifying root causes of stockouts, excess inventory, and long lines
- Comparing staffing to demand patterns
- Examining weather links to demand through simple correlation

Predictive modeling

- Simple moving average
- Weighted moving average
- Simple exponential smoothing
    - Holt Winters style seasonal logic (conceptual, not implemented in code)

These models help estimate future demand and support planning for staffing, production, and inventory management.

Prescriptive Analytics

- Decision rules for optimal staffing and inventory levels
- Strategic recommendations derived from analytics outputs

5. Evaluation

CRISP-DM emphasizes evaluating whether the models meet business objectives (Quantic, *Solving Problems with Business Analytics*, 2025). In our case, evaluation focuses on whether the Power BI outputs address the core challenges described in the stakeholder analysis. We rely on error metrics such as Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) to assess forecast accuracy.

Key evaluation questions included:

- Does the model accurately capture seasonal sales spikes?
- Are staffing recommendations aligned with historical demand patterns?
- Do the analytics insights support expansion decisions such as launching a mobile unit?

6. Deployment

Deployment in this context takes the form of a Power BI dashboard that can be used by Crafted Cones leadership (Quantic, *Solving Problems with Business Analytics*, 2025). While full CRM and ERP system integration is a future state, the dashboard acts as the first version of an analytics operational tool.

CRISP-DM notes that deployment does not need to be a technical rollout. For this project, it is the reporting and recommendation stage.

Key outputs:

- Interactive dashboard visualizations
- Executive-ready insights
- Strategic recommendations for scaling analytics maturity

Tools and Technologies

This project has selected PowerBI and Time series forecasting via Python. Explanations for the usage of these tools and technologies will be outlined below.

PowerBI

Power BI has been used for dashboard creation as it is a member of the Microsoft Suite already easily accessed by Crafted Cones.

A dashboard (illustrative example available in Appendix) was suggested to address the ongoing data needs of Crafted Cones to address the following concerns:

- Centralized Data Access: The dashboard centralizes various data points into a single platform for easy access and management.
- Customized User Views: Views are tailored to match Crafted Cones' expansion goals and operational workflows for better decision-making.
- User-Friendly Design: The platform features an intuitive and easy-to-navigate interface that enhances user experience and productivity.

The dashboard will allow Crafted Cones’ to have on-demand access to:

- Real-Time Sales Tracking and Forecasting Models: Tracks sales data instantly to provide up-to-date performance metrics and trends. View results of sales forecast models to provide strategic insights and develop policies and action plans.
- Inventory Management: Monitors stock levels dynamically to avoid shortages or excess inventory waste.
- Customer Insights: Analyzes customer data to optimize marketing strategies and product offerings effectively.

Time-Series Forecasting

Python was used to conduct predictive modeling and time-series forecasting analyses. Python provides robust statistical and machine learning capabilities that are well suited for forecasting tasks.

Given that Crafted Cones is a relatively new business with limited historical data, forecasting models were used primarily to demonstrate how predictive analytics could support future planning.

Time-series forecasting enables the business to:

- Anticipate seasonal sales patterns
- Improve staffing and inventory planning
- Support strategic decisions regarding expansion opportunities

Alignment with Project Goals and Deliverables

Goals for each of the three phases:

- **Phase 1**: Establishing the CRM and ERP System Foundation
    - Goal: Establish a unified system for capturing accurate, connected, real-time data across all business functions.
- **Phase 2**: Developing Analytics Capabilities to Support Better Decisions
    - Goal: Transform centralized data into insights that answer core business questions and guide decision-making.
- **Phase 3**: Apply Insights to Strengthen Business Performance
    - Goal: Use analytics from all four pillars to implement strategic, measurable operational improvements.

Architect the Solution: Proof of Concept

Creation of multi-data type dashboard (illustrative example available in Appendix)

Dashboard Creation

A Power BI dashboard was developed to allow the owners of Crafted Cones to interact with operational and customer analytics through a centralized visualization environment. FourSight Analytics engagement with Crafted Cones stakeholders allowed for business understanding to ensure the dashboard would reflect the needs of Crafted Cones stakeholders. The dashboard is divided into four tabs:

- Sales
- Customer
- Marketing
- Workforce

This structure allows leadership to explore performance across the core operational areas of the business.

The dashboard development process followed several steps:

Variable Identification and Data Dictionary Creation
Key variables needed by Crafted Cones were established in collaboration with stakeholders. A data dictionary was created to ensure consistent understanding of the dataset across the consulting team and business owners.

Data Digitization and Dataset Creation
Crafted Cones' historical data was converted from manual records into electronic format. Artificially generated data was then used to supplement the dataset to support the creation of a proof-of-concept analytics environment.

Data Cleaning and Transformation
The dataset was cleaned and transformed prior to visualization. This process included validating data types, correcting formatting issues, and ensuring the dataset followed third normal form where appropriate.

Dashboard Visualization Design
Visualizations were selected to ensure insights could be interpreted easily regardless of the client's level of data literacy. The dashboard presents descriptive analytics related to operations, workforce performance, and customer engagement.

Customer Sentiment Analysis
The completed dashboard was shared with the client for feedback and will be refined based on stakeholder input to ensure usability and accurate interpretation of the data.

The Customer tab of the PowerBI Dashboard included a sentiment analysis. Qualitative responses to a customer feedback survey has been included in the dashboard analytics. This is suggested to be undertaken in the future to ensure the owners of Crafted Cones understand the feedback provided by their client base and the optimal improvements that can be made to the business. At this time, responses to the customer survey were generated via artificial intelligence for demonstrative purposes at the proof of concept stage. The following actions were taken to undertake the sentiment analysis of customer responses to the customer feedback survey:

1. Artificial intelligence was used with the below prompt to generate customer responses to a customer feedback survey:

> *You are an analytics assistant helping to generate clear, realistic, and insightful customer feedback survey responses using my dataset.*

The dataset includes the following fields: Survey_ID, Customer_ID, Survey_Date, Satisfaction_Score (1–5), Comments, Comment_Sentiment, Resolution_Status, and Follow_Up_Date.

Using this schema, generate a set of customer feedback responses that represent a realistic mix of satisfaction levels, sentiments aligned with scores, and comments consistent with small ice cream shop customer experiences.

Follow these rules:

1. Scores & Sentiment Alignment
    - 5 → highly positive sentiment
    - 4 → moderately positive
    - 3 → neutral/mixed
    - 2 → moderately negative
    - 1 → strongly negative
2. Comments
    - Keep them natural, varied, and consistent with ice cream shop experiences (product quality, wait times, staff friendliness, flavor availability, store cleanliness, etc.).
3. Resolution_Status & Follow_Up_Date
    - Only include follow-up for scores 1–2, or any comment signaling dissatisfaction.
    - Follow_Up_Date should be within 3–7 days of the survey date.
4. Volume & Formatting
    - Generate N responses (I will specify the number when I run the prompt).
    - Present results as a structured table or JSON array.
    - Ensure all fields are filled and logically consistent.
5. Style
    - Keep language natural and brief but descriptive.
    - Ensure variability—no repeated comments or identical sentiment patterns.

2. Data was imported into a descriptive dashboard. Further steps of this dashboard as a whole have been outlined above.
3. Manual sentiment analysis was undertaken using DAX transformation within the PowerBI platform.
4. A manual review of the customer responses were undertaken where the data was filtered for only unique responses.

5. The following terms and requisite numbering was purposely selected by the consultant team. These terms were selected to ensure that each response was associated with only one rating. As a result, not all terms are reflective of customer sentiment[4]. 1 is considered extremely negative while 5 is considered extremely positive:

Number Assignment to Term	Terms Associated with Sentiment Score
5	Excellent, perfect, loved, friendly, exceptional, highly, absolutely, best, expand
4	Good, clean, pleasant, enjoyed, open, ambiance, seasonal, Minor, tasty, options, seating, very, truck, delicious
3	Decent, okay, acceptable, fine, crowded, flavours
2	Bland, below
1	Terrible, uninterested, poor, long, disappointed, impressed

Table 11: Terms used for manual sentiment score in PowerBI

6. The following is an example of the coding used for sentiment analysis. The variable "Customer Sentiment" contains the sentiment values based on the IF function in the following code:

```
Customer_Sentiment = IF(
        CONTAINSSTRING([Comments], "excellent") ||
        CONTAINSSTRING([Comments], "perfect") ||
        CONTAINSSTRING([Comments], "loved") ||
        CONTAINSSTRING([Comments], "Friendly") ||
        CONTAINSSTRING([Comments], "Exceptional") ||
        CONTAINSSTRING([Comments], "highly") ||
        CONTAINSSTRING([Comments], "absolutely") ||
        CONTAINSSTRING([Comments], "Best") ||
        CONTAINSSTRING([Comments], "Expand"),
        5,
```

This was repeated for each of the sentiment scores and terms outlined in table 11.

7. A new column was generated within PowerBI based on this coding in the point above.

8. Visualizations were created within the customer analytics tab of the dashboard to reflect this data, as shown in figure 8. Average and median sentiment were considered. The average sentiment and median sentiment were similar, so the average has been expressed for simplicity.

[4] The consultant team suggests using natural language processing to further undertake a sentiment analysis on active data as it would be both labour intensive and highly unlikely that the same manual process could be followed in the case of a large volume of data.

Exploratory analysis was also undertaken in conjunction with customer demographic data to further gain understanding in potential customer segments. For example, the following visualization is found in the dashboard concerning customer sentiments over time along with a customer satisfaction target.

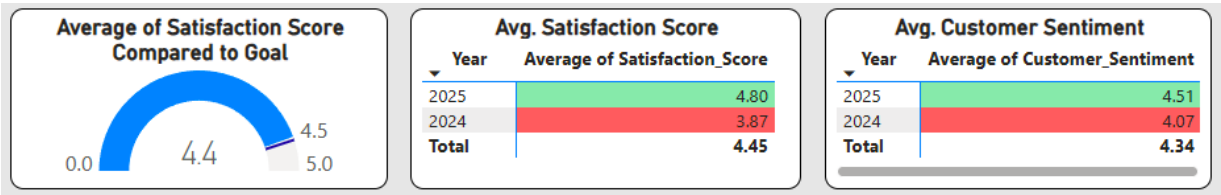

Figure 8: Customer sentiment and satisfaction against target, overtime, in PowerBI dashboard

Evaluate Business Value: Cost Analysis

Capital Requirements

The following funds will need to be allocated to implement the solution proposed by FourSight Analytics:

- Consultancy Service ($150USD/hr)
  - Completed work related to Crafted Cones = 467h (totals $70,000.00 USD)
- Enterprise Resource Planning (ERP) System[5]
  - Implementation fees: $1,500 to $15,000 USD (Source: Morrision, 2026)
  - Software fees: $1,500 to $10,000 USD (Source: Morrision, 2026)
- Customer Relationship Management (CRM) System (Source: Pricing Calculator: Zoho CRM, nd.)
  - Software fees for two user licenses: $92 USD/month

Summary of Cost Analysis

A robust cost-benefit analysis was conducted to quantify the financial impact of implementing analytics-driven operational improvements. The results demonstrated the significant value analytics can bring to Crafted Cones across workforce management, flavor production, and marketing optimization.

This section outlines cost reductions, projected ROI, and long-term value creation in detail.

Cost Benefit Analysis

A cost benefit analysis has been completed to quantify the financial impact of the proposed solution and validate the approach will result in a positive result for Crafted Cones. The framework to conduct this

[5] FourSight Analytics does not provide ERP or CRM systems for usage. These are purely estimates based on industry reports (ERP) or a cost estimate provided by a potentially contracted company (CRM). FourSight Analytics encourages you to do your due diligence in selecting an ERP and CRM system that is right for your business.

cost benefit analysis borrowed from the three phase approach proposed by FourSight analytics. These three phases are:

1. Establish a unified system for capturing accurate, connected, real-time data across all business functions (Phase 1) .
2. Transform centralized data into insights that answer core business questions and guide decision-making (Phase 2).
3. Use analytics from all four pillars to implement strategic, measurable operational improvements (Phase 3).

   The benefit to cost ratio is 1.37 indicating the three phase proposal by FourSight is a positive endeavor.[6]

[6]When building the synthetic data, we built in organic growth rates year over year which allow for a positive cost benefit ratio to be reflected. .

Cost Benefit Analysis Summary	
Year	**2025**
Total Costs	**$7,335,304.36**
Total Benefits	**$10,023,400.00**
Net Benefits	**$2,688,095.64**
Benefit-Cost Ratio (BCR)[7]:	**1.37**
Recommendation	**Proceed ✓**

Table 12: Summary of Cost Benefit Analysis

Costs[8]		
Category	**Description**	**Estimated Value (Annual)**
Direct Costs	Hardware	$1,000.00
	ERM Installation*	$7,000.00
	ERM Software	$5,500.00
	CRM Software (2 Users)	$1,104.00
	Labor	$317,124.36
	Materials	$3,141,600.00
	FourSight Consulting Fees (Proposed Project)	$70,000.00
Indirect Costs	Rent	$36,000.00
	Utilities	$10,200.00
	Insurance	$3,000.00
	Workers Compensation	$1,500.00
	NY State Disability	$700.00

[7] CBR presented is for a full business view. Considering incremental costs and benefits the CBR would be 3.61 suggesting the solution proposed by Foursight has a strong business case. ("Cost Benefit Analysis: An Introduction.", 1982).
[8] Costs were obtained as estimated values for this fictitious company using Gemini. Sources for ERP and CRM systems are noted in the *Capital Requirements* section.

	Taxes - State and Federal	$2,856,000.00
	Taxes- Property	$8,000.00
	Taxes- Sales	$821,576.00
Intangible Costs	Temporary decrease in product served while ERP/CRM system is being implemented	$12,000.00
	Employee learning curve with ERP/CRM system	$8,000.00
Opportunity Costs	Option to hire more staff to fulfill orders	$15,000.00
	Option to fix data problems internally	$20,000.00
Total Costs		$7,335,304.36

*Denotes one time cost.

Table 13: Cost Benefit Analysis, Crafted Cones Costs

Benefits[9]		
Category	**Description**	**Estimated Value (Annual)**
Direct Benefits	Gross Sales	$9,520,000.00
Indirect Benefits	Reduction in product waste	$142,800.00
	Overtime savings due to scheduling optimization	$45,000.00
Intangible Benefits	Reduction in employee turnover	$30,000.00
	Accurate promotions (Increasing loyalty conversions, decreasing reliance on ineffective discounts, etc.)	$190,400.00
	Accurate data collection to inform business insights (providing competitive advantage)	$95,200.00
Total Benefits		$10,023,400.00

Table 14: Cost Benefit Analysis, Crafted Cones Benefits

[9] Gross sales has been obtained from the dashboard provided as a deliverable of this consultancy project. The remaining benefits were obtained as estimated values for this fictitious company using Gemini.

Operational Cost Reductions

The cost benefit analysis outlined above details both the direct, indirect and intangible costs and benefits. The benefits identified provide clear opportunities to reduce annual operating expenses by optimizing scheduling, reducing turnover, and lowering waste. The following section will explore these waste reduction initiatives in more detail.

a. Overtime Reduction — $45,000 Annual Savings

   Overtime spending at Crafted Cones reached $150,000 annually, largely due to:

   - Understaffing during peak hours
   - Last-minute shift changes
   - Manual scheduling processes
   - Lack of predictive demand forecasting

   The forecast-driven scheduling model recommended:

   - Better alignment of staff to projected traffic
   - Adding shift leads to improve productivity
   - Hiring an assistant manager to improve workload distribution

   These changes can reduce overtime by approximately 30%, resulting in $45,000 in annual savings.

b. Turnover Cost Reduction — $30,000 Annual Savings

   The hospitality industry has one of the highest turnover rates (Society for Human Resource Management, 2023), with replacement costs averaging $3,500 per employee.

   Crafted Cones experienced:

   - High turnover among new and partially trained employees
   - Burnout-influenced resignations
   - Productivity losses during onboarding periods

   By implementing:

   - Standardized training
   - Assistant manager oversight
   - Improved scheduling and workforce balance
   - Employee engagement strategies (e.g., “Captain Cone” recognition program)

   Turnover can be reduced by at least 15%, equating to $30,000 in annual savings.

c. Waste Reduction — $142,800 Annual Savings

Flavor waste costs Crafted Cones approximately $714,000 ,000 annually.

Waste is driven by:

- Overproduction of seasonal flavors
- Inaccurate batch forecasting
- Inconsistent training related to production processes

Menu optimization and batch-size adjustments informed by analytics are expected to reduce waste by:

- 20%, resulting in $142,800 in annual savings.
-

d. Promotion Cost Recovery — $190,400 Annually

Crafted Cones frequently uses broad off-season discounts that:

- Reduce margins
- Do not significantly increase traffic
- Fail to boost long-term customer loyalty

With targeted marketing informed by segmentation and sentiment data, the store can:

- Decrease reliance on ineffective discounts
- Increase loyalty program conversions
- Align promotions with weather-driven demand

These improvements could recover approximately $190,400 in lost promotional value annually.

e. Accurate data collection to inform business insights (providing competitive advantage) - $95,200

Crafted Cones was previously using various means of manual tracking of data. This makes it hard for the company to make accurate and reliable business decisions informed by data. The solution proposed by Foursight has an intangible benefit of gaining competitive advantage in the marketplace, especially during the summer months when there is an increase in sales.

Total Annual Savings: $503,400

These savings will improve Crafted Cones' operational efficiency and financial stability.

Assumptions

The following assumptions underpin the financial projections provided in the cost benefit analysis:

- Sales in 2025 have increased year over year since 2023. We have provided the cost benefit analysis assuming sales are equal or better to that in 2025.
    - This assumes that both the volume of sales will remain the same, and customers will still have a positive perception of Crafted Cones.
- The cost benefit analysis assumes that $1 today is $1 in a year from now, though the time value of money theory would suggest otherwise.
- The cost benefit analysis outlines opportunity costs for alternative options to maintain the business. We assume that this method will be the best use of funds for Crafted Cones.

These assumptions were acquired through the review of historical data, as well as future looking projections. These assumptions are in line with industry standards, Srinivasan (2025) for example notes brand perception is important to foster customer loyalty and drive referrals. This will help with maintaining the volume of sales as noted in the assumption. Srinivasan (2025) also notes customer sentiment analysis is a means to further understand your customers. FourSight has already incorporated customer sentiment analysis into the dashboard, reaffirming there this assumption will be monitored in the coming months.

Looking to our second assumption, Cote (2022) and Newton (2021) note the time value of money theory posits funds have more value the sooner they are received rather than the anticipation of future funds. An assumption outlined is that $1 today is worth the same as $1 a year from now, though in reality the funds are worth less. When making monetary considerations business owners have to consider multiple factors such as inflation and perceived risk in the economy (Newton, 2021). On this note, we turn to the opportunity cost set out in assumption three. The cost benefit analysis is proposing that the method proposed by Crafted Cones is the best option to move forward, rather than other potentially less risky endeavors such as investments. Utilizing the time value of money theory, though Crafted Cones can't be certain that there are additional sources of income to continue into the future, recent studies have suggested data driven organizations " 20% increase in revenue, a 15% reduction in operational costs, and a 10% improvement in customer satisfaction compared to those that did not" (Hossain, Yasmin, Biswas, Asha, 2024).

ROI Analysis

The benefit cost ratio for this cost benefit analysis is 1.37. This suggests a positive return on investment (ROI).

Capital City Training and Consulting (2023) indicates The duration to achieve the payback period is:

Payback Period = Initial Investment / Annual Cash Inflow

In the case of Crafted Cones:

Payback Period = $83,604/$2,688,095.64
Payback Period = 0.031 years (Approximately 11 days)

As demonstrated by the payback period calculation above, it will take 11 days for Crafted Cones to pay back the initial investment. This assumes that there are no changes to the proposal which could increase or decrease the time needed to payback the initial investment.

Risk & Reward Analysis

Crafted Cones is anticipated to achieve the payback period in fewer than two weeks to undertake this initiative. As a result, Foursight Analytics predicts it is likely for Crafted Cones to have higher profit margins, increased customer reach, lower exposure to seasonal declines, stronger brand visibility, and more flexible staffing and inventory planning. To more holistically anticipate these changes to the business' operation the following risks and rewards are outlined below.

Risk: Data Ethics, Storage, and Usage

One of the primary risks associated with implementing a data-driven analytics environment involves data ethics, storage, and responsible data usage. As Crafted Cones begins collecting more structured operational and customer data—particularly through a potential customer loyalty program—the company must ensure that customer information is handled responsibly.

Improper handling or disclosure of personal data could negatively impact customer trust and damage the business's reputation. To mitigate this risk, Crafted Cones should adopt industry best practices related to data privacy and ethical data governance.

Two important ethical principles relevant to this initiative include the privacy principle and the intention principle (Quantic, *Data Ethics and Privacy*, 2025).

The privacy principle emphasizes that organizations have an ethical responsibility to ensure that collected data is securely stored and protected from unauthorized access. Customer data should not be shared with third parties unless customers are informed and provide explicit consent.

The intention principle suggests that businesses should only collect the data necessary for legitimate analytical purposes. Collecting excessive or unnecessary information increases both ethical risks and data management complexity.

To address these concerns, Crafted Cones should implement the following practices:

- Develop a clear privacy policy explaining how customer data will be collected, stored, and used.
- Provide customers with the option to opt out or request deletion of their personal data.
- Clearly define what data is required for the loyalty program and avoid collecting unnecessary information.

- Implement secure data storage protocols and restrict raw data access to authorized personnel only.

Limiting access to sensitive data reduces the risk of improper disclosure while maintaining data integrity for analytical purposes.

Reward: Potential increase in sales and strategic usage of promotional offers

Currently, Crafted Cones relies heavily on manual data collection and intuition when making operational decisions. This limits the business's ability to understand customer behavior, identify demand trends, or evaluate the effectiveness of marketing initiatives.

By implementing a centralized analytics environment, the business can shift toward data-informed decision-making. Sales analytics, customer segmentation, and sentiment analysis allow leadership to design more targeted promotions, optimize pricing strategies, and improve overall marketing efficiency.

These insights can lead to improved customer engagement and increased revenue, particularly during off-season periods where targeted promotions can be more effective than broad discounts.

Reward: Strategic usage of data to obtain a positive work culture and retention

Employee retention is a critical factor in the success of seasonal hospitality businesses. Workforce analytics can help Crafted Cones better understand patterns related to employee turnover, scheduling challenges, and training effectiveness.

For example, analyzing employee departure records or exit survey responses may reveal underlying issues related to scheduling practices, workload distribution, or workplace culture.

Using these insights, Crafted Cones can develop strategies to:

- Improve scheduling fairness
- Strengthen training and onboarding programs
- Reduce employee burnout during peak seasons
- Enhance employee recognition programs

These improvements contribute to a stronger workplace culture and increase the likelihood that seasonal employees will return in future years.

Reward: Reducing waste in both product and funds with data informed operations

Operational analytics can also reduce unnecessary costs by improving production planning and inventory management. Currently, the business produces a wide range of flavors and products to ensure availability for customers, but this approach may lead to excess inventory and ingredient waste.

Analytics-driven forecasting allows Crafted Cones to better align production with expected demand. Improved forecasting can reduce ingredient waste, optimize batch production, and improve overall inventory efficiency.

Reducing waste not only lowers operating costs but also allows the business to allocate resources toward higher-value initiatives such as marketing, staff development, and expansion planning.

Overall, as demonstrated, the benefits of utilizing a data informed business strategy outweigh the potential risks. Crafted cones should be sure to mitigate any potential risks when considering data usage, security and privacy. In conjunction with FourSight Analytics to ensure both data policies and tools are adhering to best practices.

Scale the Proof of Concept: Recommendations and Next Steps

Pilot Implementation

To transition from a proof-of-concept analytics environment to operational deployment, Crafted Cones should begin by piloting the forecasting model during the next business quarter. During this period, leadership should closely monitor both forecast accuracy and operational outcomes to determine how effectively the model supports staffing decisions, inventory planning, and seasonal demand management.

Testing the model within a controlled operational environment allows the business to refine forecasting assumptions, identify data limitations, and ensure that predictions align with real-world sales patterns before broader implementation.

CRM System Implementation

A phased implementation of a Customer Relationship Management (CRM) system should follow the pilot phase. The CRM system will allow Crafted Cones to centralize customer, sales, and marketing data, improving the quality and reliability of information used for analytics.

Key benefits of CRM implementation include:

- Improved data capture and data quality
- Enhanced customer segmentation and engagement tracking
- Integration of customer feedback and loyalty program data
- Support for long-term analytics development

This step represents a critical foundation for expanding analytics capabilities and enabling more advanced predictive modeling in the future.

Performance Measurement and KPI Development

To evaluate the effectiveness of the analytics system, Crafted Cones should establish a set of key performance indicators (KPIs) that track operational improvements over time. These metrics will allow leadership to measure the impact of analytics-driven decision-making.

Recommended KPIs include:

- Forecast accuracy to evaluate predictive model performance
- Staffing efficiency, measured through labor utilization and overtime levels
- Inventory turnover and waste reduction to monitor production efficiency
- Customer satisfaction and sentiment trends derived from survey and loyalty program data

Tracking these metrics will help ensure that the analytics framework continues to support business goals and operational improvements.

Strategic Outlook

The analytics roadmap presented in this project provides Crafted Cones with a scalable pathway from manual operations to a data-enabled business model. By implementing CRM and ERP systems, centralizing operational data, and adopting forecasting models and interactive dashboards, the business will gain improved visibility into sales trends, workforce efficiency, and customer behavior.

These insights will allow leadership to reduce operational waste, optimize staffing levels, and make more strategic marketing decisions while improving overall business performance. In addition, analytics-driven forecasting could help to support strategic decision making and inform future expansion opportunities.

As Crafted Cones continues to collect structured operational and customer data, the business will be able to refine predictive models, enhance customer segmentation, and support long-term strategic planning. Over time, this data foundation will allow the company to move toward more advanced analytics techniques, including automated forecasting, dynamic pricing strategies, and real-time operational monitoring.

Ultimately, the analytics transformation outlined in this project positions Crafted Cones to operate more efficiently, strengthen its connection with the local community, and scale its brand in a sustainable and data-informed manner.

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Appendix

GitHub Repository

A GitHub Repository was created to house all data files related to the project, including datasets, data dictionary, relational schema, illustrative customer dashboards, predictive models, stakeholder presentation slides, capstone project agreement, and individual reflective statements.

Synthetic Data

Overview of Synthetic Data Generation and Structure

This project uses a three-year synthetic dataset constructed to support analysis and demonstrate the evolution of analytics capability over time for a small, owner-operated ice cream business. While all data is simulated, the nature, granularity, and intent of the simulation differs by year to reflect realistic stages of small-business data maturity.

The dataset is organized as a relational model composed of multiple interdependent tables. Tables are linked through shared identifiers (e.g., Product_ID, Employee_ID, Customer_ID, Campaign_ID), enabling integrated analysis across operations, workforce, customers, and marketing. The interdependence of tables was intentionally designed to mirror real-world business systems and to support scalable analytics use cases.

Data Simulation Across Years

Year 1 (2023): Manual-Style Data, Simulated as a Baseline Example

Data Characterization

- Year 1 (2023) data represents manual-style operational data, similar to what a small ice cream shop might track in its earliest stages
- Although simulated, this dataset was intentionally designed to look and behave like manually maintained data

Simulation Approach

- Microsoft Copilot was used to:
  - Define table structures
  - Apply business logic
  - Generate baseline synthetic data
- Emphasizes characteristics typical of low-maturity data environments:
  - Inconsistent structure
  - Limited granularity

        - Gaps and approximations
    - Simulation prioritized:
        - Simplicity over precision
        - Aggregated or estimated values
        - Minimal automation assumptions

While simulated, the 2023 dataset intentionally mirrors the constraints and imperfections of manually tracked small-business data.

Years 2–3 (2024–2025): Fully Simulated Forward-Looking Data

Data Characterization

- Data for 2024 and 2025 was fully simulated to represent how business data could appear as processes, tooling, and analytical maturity improve over time
- These years assume:
    - Improved structure
    - Increased consistency
    - Greater analytical readiness

Simulation Approach

- Microsoft Copilot was used to:
    - Define table structures
    - Apply business logic
    - Generate baseline synthetic data
- ·Microsoft Excel randomization functions were used to:
    - Introduce natural variability
    - Simulate non-uniform behavior
    - Preserve realistic distributions

Relational Design and Table Interdependence

The synthetic dataset was designed as an interdependent relational model, rather than as isolated tables. Key design principles include:

- Shared primary and foreign keys to support joins across domains
- Consistency across years
- Logical dependencies that reflect real business workflows

Examples of table interdependence include:

- Employee_ID linking Employee_Details, Employee_Shift, Employee_Training, and Employee_Turnover tables
- Customer_ID linking Customer_Demographics, Customer_Preferences, and Customer_Satisfaction tables

- Campaign_ID linking Marketing_Tactics and Marketing_Social tables
- Product attributes designed for downstream linkage to analytical extensions

This structure enables holistic analysis across operations, workforce, customers, and marketing, consistent with enterprise analytics practices.

Sample AI Prompts Used for Data Generation

The following prompts represent the types of prompts used to generate the synthetic data contained in the data file. Prompts were iteratively refined and combined with Excel-based randomization to achieve realistic, analysis-ready datasets.

1. Overall Relational Schema Prompt

   "Generate a comprehensive relational schema for a small ice cream shop. Include tables for Products, Employees, Employee Shifts, Training, Turnover, Customers, Customer Preferences, Customer Satisfaction, Marketing Campaigns, and Marketing Engagement. Use normalized tables with unique primary keys, realistic field names, and relationships that support operational, marketing, and customer analytics. Do not include sales transaction data."

   Key Assumptions

   - Single-location ice cream shop
   - Data supports analytics, not operational execution
   - All data is synthetic and anonymized
2. Product_Details Table

   "Create a product master table for an ice cream shop. Include Product_ID, Product_Type (Cup/Cone), Product_Size (Kids, Small, Medium, Large), Number_of_Scoops, Flavor, Product_Name, and Unit_Price. Use realistic flavor names and ensure prices increase logically by size."

   Assumptions

   - Limited flavor list
   - Unique product identifiers
   - Realistic but fictional pricing

   Constraints

   - No duplicate Product_IDs
   - Prices in USD
   - All fields populated
3. Employee_Details Table

"Generate an employee master table for a retail ice cream shop. Include Employee_ID, First_Name, Last_Name, DOB, Gender, Town, State, Hire_Date, Departure_Date, Job_Title, Years_Experience, Supervisor_ID, Hourly_Wage, and Year."

Assumptions

- Roles include Supervisor, Cashier, Server
- Some employees have departure dates to simulate turnover
- Supervisor_ID references another employee

Constraints

- Valid date formats
- No duplicate Employee_IDs
- Supervisor exists in the table

4. Employee_Shift Table

"Generate a daily employee shift schedule for the main branch. Each day has two shifts: Afternoon (10:30am–5:30pm) and Evening (4:00pm–11:00pm). Include Shift_Code, Employee_ID, Date, Day, Start_Time, End_Time, Duration, Shift_Type, Branch, Month, and Year."

Assumptions

- All shifts are 7 hours
- Employees rotate across shifts

Constraints

- No employee works more than one shift per day
- Every shift is staffed
- Unique shift codes

5. Employee_Training Table

"Create an employee training table with Training_ID, Employee_ID, Course_Name, Completion_Date, and Year. Assume all employees complete Food Safety, POS System, Inventory Management, and Team Leadership training."

Constraints

- Each record linked to an employee
- Valid completion dates
- No missing fields

6. Employee_Turnover Table

"Generate employee turnover records including Employee_ID, Hire_Date, Departure_Date, Promotion_Date, Tenure_Days, Change_Type (Attrition or Promotion), Reason_For_Turnover, and Exit_Interview_Notes."

Assumptions

- Not all employees leave
- Reasons include school, relocation, or promotion

Constraints

- Tenure calculated from dates
- Reason required for each record

7. Customer_Demographics Table

"Generate fictional customer records for a NY-based ice cream shop. Include Customer_ID, First_Name, Last_Name, Full_Name, Birth_Month, Gender, City, State, Join_Date, Membership_Type, and Customer_Group."

Assumptions

- Customers reside in New York
- Cities represent Nassau County towns

Constraints

- Unique Customer_IDs
- No missing values

8. Customer_Preferences Table

"Generate synthetic customer preference data including Customer_ID, Favorite_Flavor, Serving_Type (Cone/Cup), Preferred_Channel, Discount_Sensitivity, Preferred_Notification, and a short Behavioral_Note."

Constraints

- One preference per customer
- Behavioral notes align with structured fields

9. Customer_Satisfaction Table

"Simulate customer satisfaction survey responses including Survey_ID, Customer_ID, Survey_Date, Satisfaction_Score (1–5), Comments, Comment_Sentiment, Resolution_Status, and Follow_Up_Date."

Constraints

- Sentiment aligns with score
- Follow-up only when required

10. Marketing_Tactics Table

    “Create marketing campaign records including Campaign_ID, Channel, Start_Date, End_Date, Target_Audience, Budget, Conversion_Rate, ROI, and Message_Content.”

    Constraints

    - Plausible conversion rates
    - Realistic small-business budgets

11. Marketing_Social Table

    “Generate social media engagement records tied to marketing campaigns. Include Campaign_ID, Platform, Post_ID, Engagement_Type, Engagement_Count, Sentiment_Score, Hashtags, and Post_Date.”

    Constraints

    - Engagement counts within realistic bounds
    - Sentiment scores between 0 and 1

12. Sales_Transaction Data Table

    “Generate synthetic point-of-sale transaction data for a single-location ice cream shop. Include Transaction_ID, Customer_ID, Product_ID, Transaction_Date, Quantity, Unit_Price, Total_Amount, Payment_Method, Sales_Channel, Day_of_Week, Month, Year, and Location attributes (Town, State). Ensure transactions reflect realistic purchasing behavior, including variability in quantity, pricing by product, and preferred payment methods. Maintain referential integrity with Product and Customer tables.”

    Assumptions

    - Each transaction represents a completed sale
    - Customers may appear multiple times across transactions
    - Pricing aligns with the Product_Details table
    - Sales channels include walk-in, mobile app, email, and social promotions

    Constraints

    - Unique Transaction_IDs
    - Total_Amount calculated as Quantity × Unit_Price
    - Valid foreign keys for Customer_ID and Product_ID
    - Dates fall within the defined year range

Limitations and Assumptions

While care was taken to generate realistic, analysis-ready synthetic data, several limitations and assumptions should be noted when interpreting results derived from this dataset.

Key Assumptions

- Assumed level of data availability and structure: The simulation assumes a specific level of data capture and consistency that may not fully represent the true variability or gaps present in real small-business environments, particularly in early-stage operations.
- Defined scope of data elements: Only selected operational, workforce, customer, and marketing data elements were included. The dataset does not attempt to fully represent all possible data points an ice cream business might generate.
- Business logic assumptions: Relationships between entities (e.g., customers, employees, campaigns) were based on reasonable but simplified business rules rather than observed real-world behavior.

Limitations of Randomization and Distribution

- Partial uniformity in distributions: Although randomization techniques were applied using both AI-assisted generation and Excel functions, some variables may exhibit more uniform distributions than would be expected in real-world data.
- Limited representation of extreme behaviors: Rare or highly irregular events (e.g., extreme demand spikes, unusual customer behavior, atypical staffing patterns) may be underrepresented.
- Synthetic randomness vs. true stochastic processes: Random values were generated within defined bounds and constraints, which can reduce natural skewness or clustering commonly found in real operational data.

Datasets

Crafted Cones Sales Transaction Data Part 1

Crafted Cones Sales Transaction Data Part 2

Crafted Cones Sales Transaction Data Part 3

Crafted Cones Sales Transactions Data All

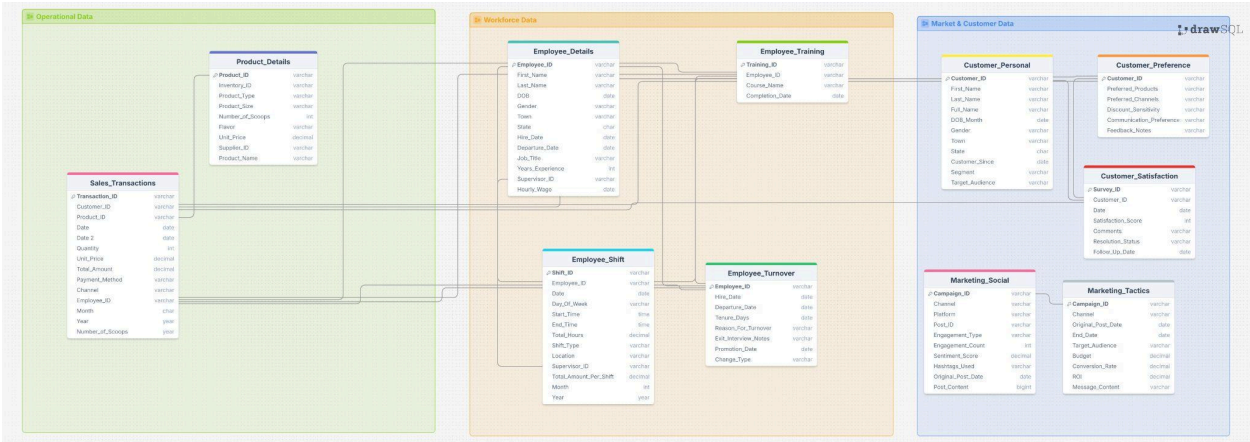
Crafted Cones Data (No Sales)

Data Dictionary

[Data Dictionary]

Relational Schema

Relational schema for proposed data project is noted below. It considers three pillars of data: Operational, workforce, as well as market and customer.



Created using Draw SQL (File can be located [here] and in [GitHub Repository]), March 6, 2026

Illustrative Dashboards

[PowerBi Dashboard File]

[PowerBi Dashboard PDF]

Predictive Model Example

[Predictive Model Example Repository]

[Predictive Model Jupyter Notebook - Example for Crafted Cones]

Stakeholder Presentation & Slides

Crafted Cones Analytics Transformation Presentation Slides

Crafted Cones Analytics Transformation Presentation Recording

Individual Reflective Statements

Each member of the team has prepared an individual reflective statement, with links to each statement provided.

H. Schroeder

B. Rai

C. Feltkamp

M. Neal

Capstone Group Project Agreement Form

Completed form